Contextual Preference Extraction from Food Reviews for Explainable Food Recommendation

1 Literature Review

Food recommender research has moved from rating- and recipe-similarity methods towards systems that use ingredients, nutrition, dietary constraints, user context and explanations [1, 2]. Extracting detailed preferences from free-text reviews is a separate problem. A single review may comment on several food aspects, mention several people, contradict itself, or state the conditions under which an opinion holds. This review surveys work in food recommendation, review-based NLP and evaluation, and positions the proposed project, *Contextual Preference Extraction from Food Reviews for Explainable Food Recommendation.*

1.1 Food preference representation and ingredient-based recommendation

Food preferences involve ingredients, taste, texture, nutrition, restrictions, price and effort, and one rating can hide the reasons behind a choice. Early work therefore related recipes to their ingredients. Freyne and Berkovsky studied personalised recipe recommendation by connecting recipe ratings to ingredient-level preferences [3]. In a follow-up UMAP study they reported that a content-based approach that breaks recipe ratings down into ingredient ratings performed best overall in accuracy and coverage [4]. A later large-scale analysis of real recipe ratings also described apparent patterns in how users reason about recipes [5]. A field survey summarises Harvey et al. as reporting better results for SVD-based factorisation than for the content and collaborative approaches proposed in [4] [1, 6]; this comparison is specific to the datasets and settings used in those studies.

The decomposition in [3, 4] treats a recipe rating as evidence about every ingredient in the recipe. This is a useful heuristic, but a rating cannot show whether the reviewer disliked the ingredient, the preparation, or the situation. The proposed project addresses this attribution problem by reading the review text instead of inferring it from a number.

Other studies model structure among recipes. Teng et al. used ingredient networks [7], van Pinxteren et al. a recipe-similarity measure for healthful meals [8], Kuo et al. sets of recipes built from ingredients [9], and Sobecki et al. hybrid recommendation in a web-based cooking assistant [10]. These show the value of recipe structure but do not say which aspects an individual prefers, or when.

Tags provide an explicit signal beyond ratings. Ge et al. combined ratings and user tags in a matrix-factorisation model and, as summarised in [1], reported significantly better prediction accuracy than content-based and standard factorisation baselines [11]. A later study found that item content contributed more to rating-prediction accuracy, whereas user tags gave better ranking quality and better perceived effectiveness in a live user study [12]. Tags are, however, short labels. They do not record who holds a preference, under what condition, or which words in the text support it. The project extends the idea of explicit preference signals by extracting these elements from free-form reviews.

1.2 User taste learning, preference attribution and context

Matrix factorisation [13] and taste-learning methods [6, 14] learn from ratings and interactions. They do not identify which person in a household holds a preference. Rostami's 2024 dissertation addresses limits of food recommenders, including reliance on past ratings, preference change over time, limited explainability, health, fairness, groups and emotions [15]. Its listed components include time-aware, explainable (deep image clustering and community detection), health- and fairness-aware, group (social community detection) and emotion-aware systems. Judging from the published abstracts, these rely on interaction data, image clustering and community structure rather than on extracting holder- or condition-level statements from review text. Demographic patterns have also been studied, for example gender differences in online cooking [16].

The closest text-based precedent in this group is Christodoulou et al., who extracted preferences and personality from review text for restaurant recommendation, using topic modelling and a transformer-based personality classifier that predicts the personality of the review's author [17]. Such features describe the reviewer. A preference voiced on behalf of someone else is not separated from the reviewer's own.

That separation matters. In "My daughter loves mild soup, but I prefer spicy soup," two preferences belong to two people. In "My partner avoids dairy, but I enjoy creamy desserts," one is a dietary restriction and the other a taste preference. Assigning all of them to the reviewer produces an inaccurate profile. Context matters in the same way: a person may like spicy food in general yet ask for mild meals when cooking for a child, or find a recipe too time-consuming on weekdays but fine on weekends. The project therefore treats holder identification and condition detection as explicit extraction tasks and keeps "unknown" as a valid value when the text is ambiguous.

1.3 Health-aware food recommendation and food choice

Elsweiler, Trattner and Harvey explored substituting typically recommended meals with similar but healthier dishes [18]. Participants could not reliably identify which of two recipes contained more fat, tended to choose the fattier one, and their choices were largely visually driven; a final user study ($n = 138$) examined whether predictive models could nudge users towards healthier recipes. The study supports keeping measured nutrition and perceived healthiness separate from taste. It addresses substitution and nudging, not extraction of preferences from reviews.

Ge, Ricci and Massimo proposed a health-aware recommender that combines a preference score with a calorie-based health score [19, 20]. Wang et al.'s Market2Dish profiles user health and recommends healthy recipes through retrieval, health profiling and recommendation components [20]. Ueta et al. built recommendation on automatically extracted nutrition information [21]. Trattner and Elsweiler analysed the healthiness of internet-sourced recipes and reported that, for the algorithms and health measures they evaluated on Allrecipes.com, recommendations tended to be unhealthy [22]. Related studies examined cooking interests and nutritional values [23], factors affecting estimated healthiness [24], and recipe upload behaviour and popularity [25, 26].

These studies show that food recommendation balances taste, nutrition, restrictions and practicality. A nutritionally suitable recipe may not suit the user's taste, and a preferred dish may conflict with a restriction. Contextual extraction can keep these requirements apart and attach each to its holder and circumstances.

1.4 Multi-aspect retrieval, knowledge graphs and language models

Recipe-MPR is a test collection for multi-aspect, preference-based natural-language retrieval of recipes [27]. Retrieval and extraction are different tasks. A query such as "a quick vegetarian dinner without mushrooms" states its constraints directly, whereas a review ("the flavor was good but the mushrooms were too strong and it took much longer than expected") expresses

them indirectly and must be interpreted first. Term weighting such as TF-IDF [28] identifies informative words but does not link a holder, aspect, sentiment and condition.

Knowledge-graph approaches represent recipes, ingredients and dietary properties. Chen et al. treated personalised food recommendation as constrained question answering over a large food knowledge graph [29]; Lei et al. used a demand-based knowledge graph for multimodal recipe recommendation [30]; DietQA combines knowledge graphs, retrieval-augmented generation and LLMs for multi-diet recipe retrieval [31]; and ChatDiet is an LLM-augmented framework for nutrition-oriented recommendation chatbots [32]. All of them depend on correctly stated user requirements. If preferences are mis-extracted or attributed to the wrong person, graph reasoning and generation inherit the error. The proposed records are an upstream input for such systems.

1.5 Datasets: Food.com, HUMMUS and FoodNexus

Food recommender research has mostly used web sources such as Allrecipes and Food.com [1]; the Food.com recipes and interactions were released for personalised recipe generation [33]. HUMMUS was collected from Food.com and contains nutrient information, text reviews and ratings, enriched with Nutri-Scores and links to FoodKG and FoodOn [34]. Bölz's thesis describes it as a food graph with about 500,000 recipes, 300,000 users and 1.9 million interactions, and reports that collaborative methods underperform on sparse data like HUMMUS, so content-based baselines, a knowledge-base question-answering system and FoodRAG variants were evaluated instead [35]. For this project that has two consequences: review text is available, and collaborative-filtering baselines should be expected to be weak.

FoodNexus (ECIR 2026 resource paper) links HUMMUS with Open Food Facts into a knowledge graph of nearly one billion triples and extracts user traits from biographies and reviews [36]. It is the closest recent resource to this project. As described in its abstract, its extraction targets user-level traits. None of these resources is described as annotating preference holders, conditions or evidence spans, which is why the project needs its own annotation protocol and challenge set.

1.6 Aspect-based and structured sentiment analysis

Aspect-based sentiment analysis (ABSA) extracts sentiment elements such as aspect term, category, opinion term and polarity, and the compound tasks extract pairs, triplets or quadruples of them [37, 38]. Structured sentiment analysis goes further and extracts full opinion tuples that include the holder, target, polar expression and polarity, formulated by Barnes et al. as dependency graph parsing over holder, target and expression spans [39], with a SemEval-2022 shared task built on it [40]. This is the nearest NLP formulation of the project's tuple. What it does not include is a condition or context element, and its benchmarks are not food-recipe reviews. The project's contribution is therefore not holder extraction in itself, but adding explicit conditions and food-specific aspects and testing whether the richer tuple prevents incorrect generalisation in user profiles.

LLMs are now a strong option for these tasks. A reality-check study evaluated LLMs on compound ABSA tasks [41], and ChatABSA, evaluated on five compound tasks over eight datasets, was reported to perform better than few-shot supervised baselines under the same few-shot settings; its comparison with fully supervised models should be read in the original paper [42]. This supports comparing dependency-parsing rules, a supervised extractor, and an LLM with a fixed output schema.

1.7 Food named-entity recognition and topic modelling

FoodBase annotates food entities [43]. TASTEset provides 700 recipes with more than 13,000 entities of types such as food, quantity, process, physical quality, purpose and taste; its best

model reached an average F1 of 0.95, ranging from 0.781 to 0.982 by type [44]. Both cover recipe text, not review text, and neither includes occasion or diner-group entities, so OCCASION, DINER_GROUP and COOKING_EQUIPMENT need new annotation. MenuNER adapts BERT to food-menu entities and was reported to perform better than an off-the-shelf BERT-base model on a handcrafted food-menu corpus drawn from customer reviews [45]. This is a precedent for the dictionary-versus-domain-adapted NER comparison.

For discovering cooking-experience themes, LDA [46], NMF [47] and BERTopic [48] are the standard candidates, and coherence measures [49] support their comparison. Because topics are statistical groupings rather than verified attributes, the project treats them as exploratory themes.

1.8 Negation, conditional preferences and behavioural testing

Negation and contrast change the preference an utterance expresses: "I dislike spicy food," "not too spicy," and "I enjoy spicy food, except at breakfast" differ in polarity, scope and exception. Negation has been annotated in many corpora, with differing definitions of scope [50]. For testing, CheckList proposes behavioural tests with minimally different inputs [51], and contrast sets probe local decision boundaries [52]. Both match the project's planned manually checked challenge set. No benchmark of conditional food preferences was identified in the sources reviewed.

1.9 Review-based explainable recommendation and evidence grounding

Review text has long been used for explanation. Explicit factor models derived explanations from phrase-level sentiment analysis [53]; NARRE provided review-level explanations [54]; and a survey organises explainable recommendation [55]. Context-aware recommender systems add situational information to the user–item model [56].

Recent LLM-based systems extract preferences from reviews directly. EXP3RT (SIGIR 2025) extracts subjective preferences from reviews, aggregates them into user and item profiles, and produces step-by-step reasoning for rating prediction [57]. MADRec (2025 preprint) builds user and item profiles by unsupervised multi-aspect extraction from reviews and uses them for re-ranking and explanation generation [58]. Shimizu et al. prompt an LLM to extract what users like and dislike from Amazon, Yelp and RateBeer reviews and keep outputs short to reduce hallucination [59]. In their abstracts, profiles are described at the user level, and none mentions separating preference holders or conditions. These are the systems against which the project's question is sharpest: does a profile that keeps holder and condition prevent a child's dislike or a weekday constraint from becoming the reviewer's permanent preference?

Explanations must also be traceable. Work on attribution in generated text [60], on generating text with citations [61], and on defining faithfulness [62] motivates the project's evidence-support and incorrect-attribution metrics. Each personalised claim should point to a review span that supports it.

1.10 Comparative synthesis and research gap

The literature shows substantial progress in each area, but the areas address different stages. Rating predictors do not identify who expressed a preference. Retrieval systems satisfy a query but do not extract persistent or conditional preferences from past reviews. Explanation generators do not by themselves guarantee that each claim is supported by review evidence. Structured sentiment analysis supplies holders but not conditions, and recent LLM-based recommenders supply review-derived profiles but, as described, at the user level. The gap addressed here is a representation, a benchmark and a test: structured records that preserve holder, food aspect, sentiment, condition and evidence span; an annotated benchmark for them; and an evaluation of whether keeping holder and condition improves profile accuracy, recommendation and explanation.

Table 1: Comparison of work families against the proposed project.

Work family	**Aspects**	**Preference holder**	**Condition / context**	**Evidence span**	**Used for recommendation**
Ingredient decomposition [3, 4]	Ingredients	No	No	No	Yes
Tag-based factorisation [11, 12]	Tags	No	No	Label only	Yes
Health-aware systems [18–20]	Nutrition	No	No	No	Yes
Rostami [15]	Health, time	Group-level	Time-aware	No	Yes
Restaurant review preferences [17]	Topics, personality	Author	No	No	Yes
FoodNexus [36]	User traits	User-level	Not described	Not described	Evaluation resource
Structured sentiment [39, 40]	Target, expression	Yes	No	Spans	No
LLM review-based recommenders [57–59]	Aspects	User-level	No	Partly	Yes
Proposed project	Food aspects	Yes, or "unknown"	Yes	Yes	Yes

Entries reflect the abstracts or descriptions of the cited work and should be confirmed against the full texts.

The intended record fields are:

1. **Preference holder:** the person whose preference is expressed, or "unknown."
2. **Food aspect:** ingredient, taste, texture, preparation effort, time, availability, perceived healthiness or another feature.
3. **Sentiment:** positive, negative, mixed or qualified.
4. **Context or condition:** when or for whom the preference applies, if stated.
5. **Evidence span:** the text supporting the record.

For example, "I enjoy salmon for dinner, but I find it too heavy for breakfast" yields a positive record in a dinner context and a negative one in a breakfast context, which a single rating would merge.

1.11 Implications for evaluation

Extraction and recommendation should be evaluated separately. Extraction can be scored with entity-level, field-level and tuple-level precision, recall and F1, with agreement reported on a double-annotated subset [63]. Recommendation experiments should use chronological splits and only historical reviews, report ranking metrics, and test contextual matching with manually authored requests and independently judged candidates, since Food.com provides no context for future requests. Explanations should be scored for evidence support and incorrect attribution. Because collaborative filtering is weak on sparse HUMMUS data [35], the comparison should include content-based, popularity, review-embedding and context-independent profile baselines built from the same extracted facts.

In summary, the project builds on food-recommendation research while focusing on a distinct extraction problem. By preserving holder, aspect, sentiment, condition and evidence, it aims to give more accurate user profiles and more transparent, context-aware recommendations.

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